Do specialized components learn their intended roles—and does this improve forecast uncertainty?

DecoVAE-FM

01 Probabilistic forecasting: task and model

Forecasting task

$\mathbf{x} = (x_{t-T+1}, \dots, x_t) \in \mathbb{R}^T$

$\mathbf{y} = (x_{t+1}, \dots, x_{t+H}) \in \mathbb{R}^H$

T: context length; H: forecast horizon.

$\mathbf{x} \longmapsto p_\theta(\mathbf{y} \mid \mathbf{x})$

Quantiles and intervals describe forecast uncertainty.

Example: 96 past hours, a 24-hour forecast.

How DecoVAE-FM represents uncertainty

Trend and seasonal outputs form the mean. Residual features also inform the variance.

$\mu = \tau + s, \qquad \log \sigma^2 = g_\theta(\mathbf{z})$

τ, s: trend / seasonal forecasts.

z collects learned feature vectors of the trend, seasonal and residual components, obtained through their encoders.

Mean and variance are in normalized model space.

02 Core models: assumptions, training and published protocols

MODEL	STRUCTURAL ASSUMPTION	TRAINING / OUTPUT	RECONSTRUCTED EVALUATION
DecoVAE [1]	Smooth trend + Fourier seasonality; two specialized VAE streams.	**Dataset-specific** MSE + KL + smoothness Latent forecast samples	7 datasets; T=96, H=24/720; 3 retrains; CRPS / NMAE. No direct calibration test.
K²VAE [2]	Linear dynamics in a lifted Koopman space; Kalman correction.	**Dataset-specific** Koopman–Kalman VAE Gaussian likelihood	ProbTS: 8 short / 9 long settings; typical T=H=24 or T=96, H≤720; 5 runs; CRPS / NMAE.
Chronos-2 [3]	Time / group attention; cross-series information and covariates.	**Pretrained / zero-shot** Direct multi-step forecast 21 quantiles	fev-bench, GIFT-Eval, Chronos-II; rolling origins; WQL / MASE and aggregate scores.
Moirai 2.0 [4]	Channel-independent decoder Transformer; multi-token prediction.	**Pretrained / zero-shot** Autoregressive quantiles 9 levels: 0.1–0.9	GIFT-Eval: 97 tasks / 55 datasets; normalized MASE / quantile CRPS; GIFT train in pretraining.

03 Additional lenses: sampling-rate transfer and calibration

WORK / PURPOSE	METHOD AND EVIDENCE	WHAT IT ADDS TO OUR PROJECT
FlowState [5] **Sampling-rate transfer**	Pretrained S5 + functional Legendre decoder; quantiles. GIFT: 97 tasks; normalized MASE / quantile CRPS; rate stress tests.	A separate test of time-scale assumptions. Sampling-rate robustness does not establish interval calibration.
Beyond Accuracy [6] **Calibration study**	Tests quantile calibration, interval coverage and width across prediction heads and forecast horizons.	Guides our calibration tests: nominal vs observed coverage, paired with interval width. Adds evidence beyond CRPS / WQL.

04 Synthesis: claims, evidence and missing tests

Accurate forecasts ≠ correct components

Compare predicted trend and seasonality with known synthetic components. Overall forecast error cannot show what each branch learned.

Good scores ≠ calibrated intervals

An 80% interval should cover about 80% of observations. Check coverage and width together: wide intervals can cover more yet tell us less.

Zero-shot ≠ unseen-data transfer

Some GIFT training data are included in Chronos-2 / Moirai pretraining. Check excluded sources before claiming unseen-data transfer.

05 Project questions

Do the branches learn their intended components?

Does residual information improve uncertainty?

[1] DecoVAE: 2608.20052 [2] K²VAE: 2505.23017 [3] Chronos-2: 2510.15821 [4] Moirai 2.0: 2511.11698
[5] FlowState: 2508.05287 [6] Beyond Accuracy: 2510.16060.